Hussein Ziyard

BSc Applied Data Science Communication (Reading)
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Data Science undergraduate skilled in Python, R, SQL, and Power BI, with a strong foundation in machine learning and analytics. Proven leader through 2+ years as an Assistant Basketball Coach. Seeking an internship to apply analytical skills to real-world data challenges.

Education

General Sir John Kotelawela Defence University — Jan 2024 – Present
Bachelor's Degree – Applied Data Science Communication

National Institute of Sports Science — Nov 2023 – Dec 2023
Certificate in Sports Science – Basketball Level 1

Zahira College — Feb 2009 – Dec 2022
Local A-Levels – Combined Mathematics, Physics, ICT

Research Papers

Predicting USD/LKR Exchange Rate: A CRISP-DM ML Approach — 2026
Presented – ADScAI 2026, University of Moratuwa — *Python, XGBoost, Pandas, NumPy*
Built 4 ML models (Random Forest, XGBoost, Gradient Boosting, Ensemble) on 16 years of LKR/USD data; XGBoost achieved MAPE of 0.838% and R^2 of 0.904 with 30-day recursive forecasting and 95% confidence intervals

EV Adoption Impact Analysis — 2025
Research Paper — *Data Analytics*
Proposed a global analytical model to study environmental and economic impacts of EV adoption across multiple economies

Academic Projects

Olympics 2028 Medal Prediction & Dashboard — 2026
Machine Learning & Data Visualisation — *Python, scikit-learn, Power BI, Excel*
Ensemble ML model (Random Forest + Gradient Boosting) on 271K+ athlete records (1896–2016) to predict per-country medals for LA 2028; R^2 of 0.917 and MAE of 3.46 on held-out 2016 Olympics; engineered lag, rolling, and trend features across 205 countries with 95% confidence intervals; interactive Power BI dashboard with 6 chart types on 3,696 participations and 22K+ medals

PolicySnap Studio – AI Policy Summarizer & Draft Generator — 2026
Academic Assignment – Intelligent Systems — *Python, Streamlit, OpenRouter API, pdfplumber*
Streamlit app that ingests PDF/TXT policy documents, calls an LLM via OpenRouter API to generate structured summaries, and produces scenario-adapted policy drafts (preset + custom scenarios)

Consumer Behavior, Patient Outcomes & Public Sector Classification — 2024 – 2025
Group Academic Project – Data Mining — *Python, R, Shiny, Scikit-learn*
Apriori Association Rule Mining on 500K+ transaction UK retail dataset; Logistic Regression for 30-day hospital readmission prediction on diabetic patient data, evaluated with ROC/AUC; EDA and CRISP-DM clustering on public sector datasets

Advanced SQL & Cloud Databases: Government & SDG Data — 2025
Academic Project — *T-SQL, Azure SQL, Power BI, DAX*
End-to-end analytics pipeline on UN SDG datasets (17 Goals, 100+ countries, 2000–2025); Power BI star schema with 5 DAX measures and 5 dashboard pages

SQL & Power BI: Government & Immigration Analytics — 2024
SQL Projects – Australian Government Data & Canadian Immigration — *SQL Server, T-SQL, Power BI, DAX*
Relational databases and complex T-SQL queries on Australian government datasets; modelled large-scale Canadian immigration data with Power BI reporting for policy trend analysis

Events & Leadership

Co-Chairperson – Data Odyssey 2025 — October 2025
AI and Data Science Club, General Sir John Kotelawela Defence University
Co-chaired inter-faculty and inter-university data science exhibition; managed logistics for 50+ participants across multiple faculties

Co-Event Organizer – Data Science Exhibitions & Competitions — Jan 2025 – Feb 2026
AI and Data Science Club, General Sir John Kotelawela Defence University
Co-organized 3+ club events including inter-faculty data science workshops and exhibitions

Chairperson – Inter-House Basketball Tournament July 2021
Zahira College Colombo
Organised and chaired annual tournament, overseeing scheduling, refereeing, and event management

Experience

Assistant Basketball Coach Jun 2023 – Present
Zahira College
Assisted training, performance analysis, and team coordination for 15+ players over 2+ years

Mathematics Teacher (Volunteer) May 2023 – Sep 2023
Al Hikma College
Delivered lessons and academic support to 20+ students over 5 months, improving comprehension and exam readiness

Technical & Skills

Programming: Python, R, SQL
ML & Algorithms: Random Forest, XGBoost, Gradient Boosting, Logistic Regression, Apriori, Clustering, Time-Series Forecasting, CRISP-DM
Libraries & Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, Streamlit, Shiny, arules, caret, pROC, ggplot2
Database & BI: SQL Server, Azure SQL, T-SQL, Power BI, DAX, Star Schema, ETL
Tools: Git, GitHub, VS Code, Jupyter Notebook, Excel
Soft Skills: Leadership, Teamwork, Analytical Thinking, Problem-Solving

Certifications

Research Paper Presenter – ADScAI 2026 – Applied Data Science & AI Conference, University of Moratuwa, April 2026
Workshop Participation – ADScAI 2026 – *Event-Driven AI on AWS (Emojot); Shipping Real Features with AI Agents (Cut+Dry)*, April 2026
Databases and SQL for Data Science with Python – IBM, February 2026
Microsoft Power Up Program – Certified Graduate
Python for Beginners – CODL, University of Moratuwa, February 2026
Certificate in Sports Science – Basketball Level 1 – National Institute of Sports Science, Sri Lanka, March 2024

Honors & Awards

Research Paper Presenter – ADScAI 2026 2026
Department of Computer Science and Engineering, University of Moratuwa

Workshop Participation – ADScAI 2026 2026
Department of Computer Science and Engineering, University of Moratuwa

Champions – Data Odyssey Inter Faculty Exhibition Competition 2025 2025
AI and Data Science Club, General Sir John Kotelawela Defence University

KDU Merit Award for Basketball – Colours Night 2025 2025
General Sir John Kotelawela Defence University

Champions – Green Basketball Championship 2025 2025
NSBM Basketball Club, NSBM Green University

Participant – Data Odyssey Inter-University Exhibition Competition 2024 2024
AI and Data Science Club, General Sir John Kotelawela Defence University

Best Basketball Player – Annual Prize Day 2022
Zahira College

University Entrance Prize – Annual Prize Day 2022
Zahira College

References

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